Keeping Alternatives Alive: A Comparison-Preserving Scaffold for Root Cause Analysis Under Closure Pressure

SORA KIM, Pharmaceutical Manufacturing Industry, South Korea

Investigations in pharmaceutical manufacturing routinely converge on individual blame before alternatives have been meaningfully compared — not because investigators reason poorly, but because the workflow does not sustain alternatives in a comparable state long enough for systematic evaluation. This paper reframes premature causal convergence as an interaction design problem: the failure is not in generating explanations, but in maintaining them under conditions that structurally collapse comparison before closure.

I introduce Premature Accountability Convergence (PAC) as a structural concept locating this failure not in individual cognition but in the interaction between closure pressure, documentation requirements, and the institutional incentives that shape what explanations remain comparable. Building on this framework, I introduce two design contributions: *comparison preservation* as a design principle for AI-supported reasoning in regulated environments, and *AI as comparison enforcer* as an inversion of the dominant model in which AI accelerates convergence by providing anchoring explanations. Where conventional AI systems narrow the comparison space by suggesting candidate causes, a comparison enforcer detects premature convergence signals and requires their reopening — sustaining comparability rather than resolving it.

I report a two-study empirical program with $N = 20$ GMP-experienced participants. Study 1 establishes the motivation: using a timed reflection protocol at three timepoints (T0, T+5, T+10), I show that convergence is time-sensitive and reflects institutional pressure rather than epistemic certainty. At T0, 15 of 20 participants produced single-cause, person-centered explanations; confidence ratings averaged 1 out of 5 across all participants regardless of experience level. Post-task discussion revealed that all 20 participants selected explanations oriented toward *documentation acceptability* rather than investigative completeness — a rational response to an accountability environment that rewards closure over thoroughness. Study 2 introduces an AI-supported RCA Reasoning Scaffolder that operationalizes the comparison enforcer design: a five-stage Streamlit application in which an AI component integrated into Stage 3 detects convergence-oriented language and triggers reopening prompts, without suggesting causes or generating conclusions. Consistent with the cognitive forcing function literature [2], the AI component functions as a structured pause rather than a recommendation engine — extended here to target not over-reliance on AI outputs but premature reliance on the investigator's own first account. 18 of 20 participants produced broader causal accounts than in Study 1; 15 of 20 reconsidered their initial attribution, with 12 of those 15 shifting from person-centered to systemic or procedural explanations. Usability ratings averaged 5.0 out of 5. All 20 participants expressed adoption intent while simultaneously identifying a structural barrier: deploying a comparison-preserving scaffold produces an audit trail documenting which alternatives were considered and not pursued — making the decision not to investigate a systemic cause an accountable act rather than an invisible one.

I argue that AI in accountability-critical environments should be designed not to accelerate conclusions but to preserve the conditions under which human judgment remains comparative — and that the organizational resistance to this transparency is itself a finding about the accountability structures that make premature convergence rational.

Additional Key Words and Phrases: root cause analysis, accountability, closure pressure, human-AI interaction, comparison preservation, regulated environments, premature accountability convergence

1 Introduction

Root cause analysis (RCA) in pharmaceutical manufacturing is a formal quality requirement embedded in collaborative investigative work: every deviation must be investigated, its cause documented, and corrective actions assigned before the record can be closed. In principle, investigation should compare multiple plausible causes before converging on an explanation. In practice, investigations unfold under closure pressure — temporal deadlines, documentation

Author's Contact Information: Sora Kim, Pharmaceutical Manufacturing Industry, South Korea, sorakim.lab@gmail.com.

requirements, and institutional expectations that define when an investigation must be considered complete. Under these conditions, the first viable explanation tends to become the documented one, not because alternatives have been eliminated, but because the workflow does not keep them comparable long enough to be evaluated.

Prior work has largely framed premature convergence as a cognitive problem: investigators rely on recognition-primed decision making [3], anchor on initial hypotheses, or fail to generate sufficient alternatives. I argue this framing misidentifies the locus of failure. Premature convergence in regulated investigations is not primarily a property of how individuals reason — it is a property of how investigative workflows organize, interrupt, and foreclose comparison under institutional pressure.

This paper introduces *comparison preservation* as a design principle for AI-supported reasoning in accountability-critical environments, and *AI as comparison enforcer* as an inversion of the dominant model for AI in investigation support. Where existing systems provide candidate explanations that anchor and narrow the comparison space, a comparison enforcer detects when comparison has collapsed and requires its reopening — acting against the structural tendency toward premature convergence rather than reinforcing it.

This paper makes three contributions. First, it reframes premature convergence as a failure of investigative system design rather than individual cognition. Second, it introduces the *comparison enforcer* as a design role for AI in collaborative investigative settings — distinct from suggestion engines or automated classifiers. Third, it identifies organizational transparency as an adoption barrier specific to regulated environments, where making reasoning visible surfaces accountability conditions that existing practice leaves invisible.

2 Related Work

Vaughan's account of the Challenger launch decision establishes that causal explanations in high-stakes organizations reflect the conditions of their production [7]. Klein's account of recognition-primed decision making shows that experts under time pressure do not compare alternatives but recognize and act on the first viable option [3] — a cognitive strategy that, in regulated investigation contexts, produces premature convergence precisely when accountability stakes are highest. Star and Strauss show that coordination work performed in the gaps of formal systems becomes invisible [5]: in regulated investigation workflows, the comparative work of holding multiple causal accounts in parallel is unrewarded by closure expectations and absent from final investigation records. Suchman establishes that action is shaped by the immediate conditions of its execution [6], implying that investigation quality is shaped not only by investigator expertise but by the visibility conditions under which reasoning unfolds.

Amershi et al. formalize guidelines for human-AI interaction emphasizing that AI systems should support human control and avoid pre-empting judgment [1]. In regulated environments, this principle takes on particular force: quality decisions must remain with qualified human reviewers. Buçinca et al. demonstrate that cognitive forcing functions — which require users to engage more deliberately with AI outputs before accepting them — reduce over-reliance without sacrificing efficiency [2]. The comparison enforcer extends this principle to a context in which the risk is not over-reliance on AI outputs but premature reliance on the investigator's own first account. This paper reframes the role of AI from suggesting explanations to enforcing the sustained comparability of alternatives under closure pressure.

3 Method

Both studies were conducted in a pharmaceutical manufacturing quality system operating under GMP regulations, with $N = 20$ participants recruited from a single pharmaceutical quality team with direct experience in deviation investigation, root cause analysis, and CAPA documentation. Participants represented varied levels of experience and

roles within the investigative workflow. This sample size is consistent with established think-aloud methodology [4] and reflects the structural constraints of conducting research within a regulated organizational setting. Both studies used the same standardized deviation scenario — an out-of-specification analytical result with sufficient contextual detail to support multiple plausible causal interpretations, including person-centered, procedural, equipment-related, and organizational accounts.

Study 1: Timed Reflection Protocol. Participants produced causal explanations suitable for formal documentation at three timepoints: immediately after reading the scenario (T0), after five minutes of unstructured reflection (T+5), and after ten minutes (T+10). At each timepoint, participants recorded their causal account and rated confidence on a 1–5 scale. Responses were analyzed for causal breadth, causal type (person-centered, procedural, systemic, or mixed), and evidence of comparison across alternatives.

Study 2: Comparison-Preserving Scaffold. Participants completed the same scenario using an AI-supported RCA Reasoning Scaffolder implemented as a Streamlit web application. The scaffold organized investigation into five stages — case intake, hypothesis expansion, PAC detection, structured comparison, and pre-closure verification — enforcing explicit comparison before closure. The AI component was integrated exclusively into Stage 3: when convergence-oriented language or single-cause fixation was detected, it surfaced alternative causal pathways and generated reopening prompts, without suggesting causes or generating conclusions. Post-task measures included causal breadth, evidence of explicit comparison, and interaction traces; post-task discussion probed organizational barriers to deployment.

4 Findings

4.1 Study 1: Convergence is Time-Sensitive and Institutionally Produced

At T0, 15 of 20 participants produced single-cause, person-centered explanations. Confidence ratings averaged 1 out of 5 across all participants regardless of experience level — establishing that early convergence does not reflect epistemic certainty but institutional pressure to produce a documented explanation quickly. With additional reflection time, explanations expanded: by T+10, 18 of 20 participants had either broadened their account or become explicitly uncertain about the initial cause. Comparison is time-sensitive; its collapse is not a fixed property of how investigators reason but a property of how much time the workflow allows reasoning to develop.

Post-task discussion revealed a finding more consequential than the timepoint data alone: all 20 participants described selecting explanations oriented toward *documentation acceptability* rather than investigative completeness — choosing contained, person-centered accounts that would satisfy audit review without opening systemic lines of inquiry. Premature convergence is therefore not a reasoning failure but a rational strategy: an adaptive response to an accountability environment that rewards closure over thoroughness.

4.2 Study 2: Effects of the Comparison-Preserving Scaffold

Under the scaffold, 18 of 20 participants produced broader causal accounts than in Study 1. The PAC detection layer triggered reopening prompts when convergence signals appeared, and 15 of 20 participants reconsidered their initial attribution; of these, 12 shifted from person-centered accounts to systemic or procedural explanations — moving from accounts attributing the deviation to an individual actor to accounts identifying the procedural or organizational conditions that made that action possible. Usability ratings averaged 5.0 out of 5; the adoption barrier was not interactional but organizational. All 20 participants expressed adoption intent while simultaneously identifying the

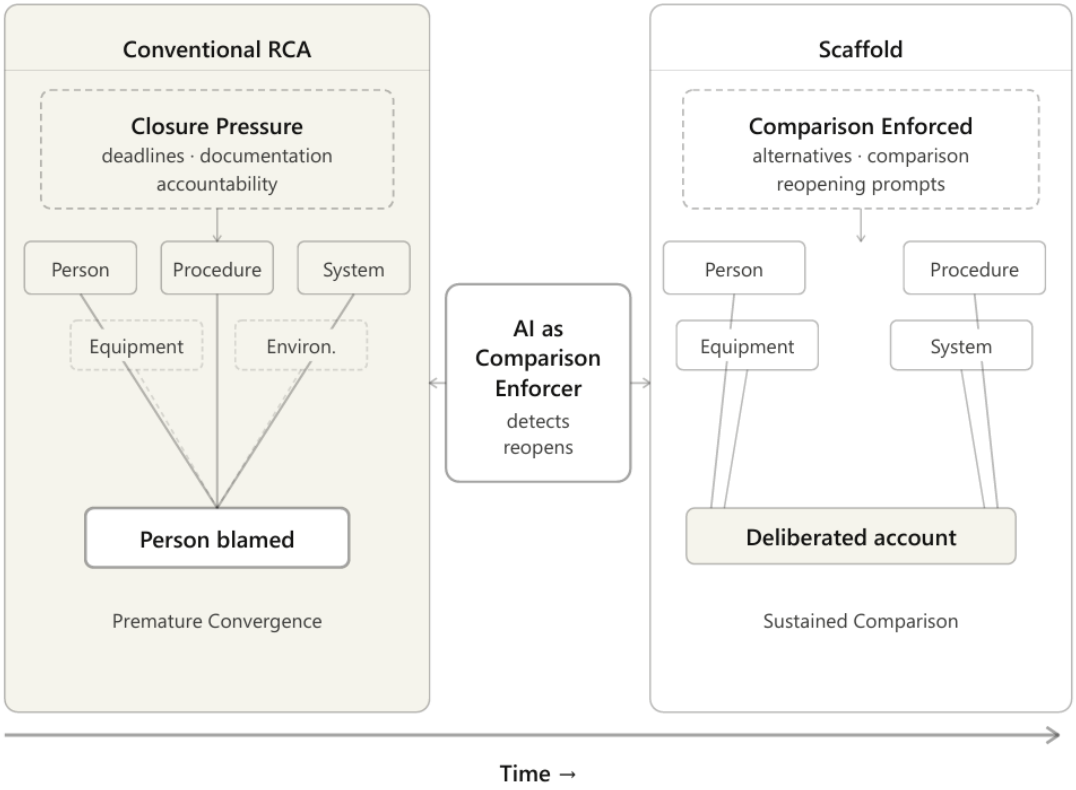

Fig. 1. Premature Accountability Convergence (left) versus a comparison-preserving scaffold (right). Under closure pressure, conventional RCA converges early, collapsing the comparison space before alternatives can be meaningfully evaluated. The scaffold preserves this space by sustaining comparison across multiple alternatives prior to closure.

audit trail as a structural barrier: when investigation reasoning is documented, the decision not to pursue a systemic explanation becomes an accountable act rather than an invisible one. As one participant noted: *"the problem isn't the tool. The problem is that the tool would show that we looked at the systemic cause and decided not to pursue it. That's a conversation no one wants to have."*

5 Discussion

The findings establish that premature convergence is a property of investigative systems rather than individual reasoning — and that it is time-sensitive, and therefore designable. The *comparison enforcer* reframes the role of AI in accountability-critical environments: not to accelerate conclusions but to sustain comparability before attribution. This inverts the dominant model for AI in RCA support, in which suggestion engines and automated classifiers provide anchoring explanations that narrow the comparison space before investigators have explored alternatives independently.

The organizational adoption barrier reveals a deeper tension: making reasoning visible may conflict with existing accountability practices that favor stable, defensible conclusions. A regulated organization that deploys a comparison-preserving scaffold must also accept that its investigation records will document which alternatives were available

and not chosen — surfacing conditions that current practice leaves invisible. Improving investigative reasoning may therefore require redesigning not only tools, but the accountability structures that govern how reasoning is evaluated.

6 Conclusion

Premature convergence in regulated investigation is not a reasoning failure but a product of how investigative systems structure comparison under closure pressure. An AI-supported scaffold that enforces explicit comparison before closure — functioning as a comparison enforcer rather than an explanation source — can expand causal breadth and delay premature attribution without automating conclusions or displacing human judgment. The challenge is not to improve AI accuracy, but to preserve the conditions under which human judgment remains comparative. This requires engaging with the organizational structures that currently make premature convergence rational — a design challenge as much organizational as it is technical.

References

[1] Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., and Horvitz, E. 2019. Guidelines for human-AI interaction. In *Proc. CHI '19*. ACM.

[2] Buçinca, Z., Malaya, M. B., and Gajos, K. Z. 2021. To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision making. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1, 1–21.

[3] Klein, G. A. 1998. *Sources of Power: How People Make Decisions*. MIT Press.

[4] Nielsen, J. 1994. *Usability Engineering*. Morgan Kaufmann.

[5] Star, S. L. and Strauss, A. 1999. Layers of silence, arenas of voice. *CSCW* 8, 1–2, 9–30.

[6] Suchman, L. A. 1987. *Plans and Situated Actions*. Cambridge University Press.

[7] Vaughan, D. 1996. *The Challenger Launch Decision*. University of Chicago Press.